PAPER_454 — MUGE Compression Cycle 2: 19-System Multi-Registry Expanded Gravitational Calculator

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Classification: FIRST 19-system UQFF/MUGE compression registry; FIRST multi-class environmental compression from magnetar through cosmological scales
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Abstract

Compression Cycle 2 expands the 7-system canonical registry (PAPER_452) to a 19-system multi-scale catalogue incorporating objects from neutron star surfaces (r~10 m) to cosmological filaments (r~10^2 m). The 12 newly added systems span HII regions, galaxy mergers, galaxy clusters, and the Hubble Ultra-Deep Field, each contributing system-specific F_env terms. The single compressed equation g_UQFF = g_Newton × (1 + H_z t) + F_env(t) applies uniformly across 22 orders of magnitude in spatial scale — demonstrating the universality of the MUGE compression framework for all observed astrophysical environments.

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2. 19-System Registry — PAPER_454

2.1 Full System Catalog

The 19-system registry extends the 7-system base (PAPER_452) with 12 additional systems:

#	System	Type	M (kg)	r (m)	F_env dominant
1	MagnetarSGR1745	Neutron star	5.58×10^3	1×10	B-field + decay
2	SagittariusA	SMBH	8.17×10^3	6×10	Accretion disk
3	TapestryStarbirth	SF region	9.96×10^{33}	1×10^1	SFR + outflow
4	Westerlund2	Star cluster	1.99×10^3	6×10^1	Stellar wind
5	PillarsCreation	HII pillars	3.98×10^{32}	6×10^1	Radiation
6	RingsRelativity	GL system	1×10^3	1×10^2	Lensing
7	UniverseGuide	Cosmological	$1\times10\ ^3$	4.4×10^2	F_cosmo
8	**NGC2525**	Barred spiral	~$2\times10\ ^1$	~5×10^2	Tidal arm
9	**NGC3603**	Massive cluster	~4×10^3	~3×10^1	OB wind
10	**BubbleNebula**	Wind bubble	~1×10^{32}	~2×10^1	Stellar bubble
11	**AntennaeGalaxies**	Merger	~4×10	~2×10^{21}	Tidal merger

#	System	Type	M (kg)	r (m)	F_env dominant
12	**HorseheadNebula**	Barnard 33	~5×10^{31}	~5×10^{1}	B-field + PDR
13	**NGC1275**	Perseus cluster	~1×10 ^{3}	~3×10^{22}	Cluster ICM
14	**NGC1792**	Spiral galaxy	~1×10 ^{1}	~4×10^{2}	Disk wind
15	**HubbleUDF**	Deep field	~10 ^{3}	~4×10^{2}	Statistical ensemble
16	**StudentsGuideUniverse**	Pedagogical	variable	variable	All terms
17	**LagoonNebula**	M8 HII region	~5×10^{33}	~1×10^{1}	Radiation + ionisation
18	**TrifiidNebula**	M20 trifurcated	~1×10^{33}	~8×10^{1}	Radiation + dust
19	**OmegaNebula**	M17 Swan	~3×10^{33}	~1×10^{1}	O-star radiation

2.2 Compressed MUGE Equation (Universal Form)

$$g_{\mathrm{UQFF}}^{(j)}(t) = \underbrace{\frac{GM_j}{r_j^2}(1 + H_z t)(1 - B_j/B_{\mathrm{crit}})}_{g_{\mathrm{Newton,Hubble,B}}} + \underbrace{\sum_k U_{gk}^{(j)}}_{U_{g1}+U_{g2}+U'_{g3}+U_{g4}} + \underbrace{F_{\mathrm{env}}^{(j)}(t)}_{\text{system-specific}}$$

2.3 New F_env Types (Systems 8–19)

Tidal merger (Antennae Galaxies):

$$F_{\mathrm{env,tidal}} = \frac{GM_1 M_2}{(d_{12})^3} \cdot r$$

Where d_{12} = separation between merging cores, r = field evaluation point.

ICM (NGC 1275 Perseus cluster):

$$F_{\mathrm{env,ICM}} = \frac{kT_{\mathrm{ICM}}}{\mu m_H r_{\mathrm{cool}}} = \frac{1.38 \times 10^{-23} \times 5 \times 10^7}{0.62 \times 1.67 \times 10^{-27} \times 3 \times 10^{22}} \approx 2.7 \times 10^{-12}\ \mathrm{m/s^2}$$

With T_ICM   5×10 K (Perseus cooling flow).

Stellar wind bubble (NGC 3603/Bubble Nebula):

$$F_{\mathrm{env,bubble}} = \frac{\dot{M}_{\mathrm{wind}} v_{\mathrm{wind}}}{4\pi r_{\mathrm{bubble}}^2}$$

OB stellar wind (Lagoon/Trifid/Omega):

$$F_{\mathrm{env,OB}} = \frac{L_{\mathrm{OB}}}{4\pi r^2 c} = P_{\mathrm{rad,OB}}$$

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3. Scale Universality of MUGE Compression

The 19 systems span 22 orders of magnitude in radius and 22 orders in mass:

Property	Min	Max	Range
Radius (m)	10 (magnetar)	4.4×10^{2} (universe)	22 dex
Mass (kg)	5.6×10^{3} (magnetar)	$10\ ^{3}$ (universe)	22+ dex
g_UQFF (m/s^2)	~$10\ ^{3}$ (ultra-low)	~10^{12} (magnetar surface)	46 dex

The **single compressed equation** handles the full range with only F_env changing between systems. This is the UQFF universality principle: **the gravitational equation is scale-free**.

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4. Total System Gravity Budget at t=1 Gyr

$$G_{\text{total}}^{(19)} = \sum_{j=1}^{19} g_{\text{UQFF}}^{(j)}$$

Dominated by Magnetar surface gravity:

$$G_{\text{total}}^{(19)} \approx 3.73 \times 10^{6} \text{ (Magnetar)} + O(10^{0 \text{ to } 2}) \text{ (others)} \approx 3.73 \times 10^{6} \text{ m/s}^{2}$$

For normalised comparison (setting g_Magnetar = 1 reference):

System	g / g_Magnetar
Magnetar	1.0
SgrA* (at r=6×10 m)	3.0×10
Galaxy clusters	~$10\ ^{1}$
Universe	~$10\ ^{21}$

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5. Standard Model Comparison

Feature	SM	CC2-19 System
Multi-class gravity	Separate codes per system	Unified registry
Scale coverage	Different equations at different scales	Single g_UQFF for 22 dex

Feature	SM	CC2-19 System
ICM coupling	Separate X-ray / hydro	F_env,ICM built-in
Tidal mergers	N-body codes	F_env,tidal in registry

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6. Testable Predictions

1. **Scale-free universality:** g_UQFF GM/r^2 for all 19 systems to leading order, with F_env/g_Newton < 10^2 for all systems — testable by comparing UQFF output at each system's characteristic radius.
2. **ICM temperature coupling:** F_env,ICM T_ICM — doubling Perseus cluster temperature to 10 K should double the F_env contribution. Testable via Chandra X-ray spectra.
3. **Tidal merger timing:** F_env,tidal for Antennae grows as d decreases — UQFF predicts F_env d ^{3} increasing by 8× as separation halves. Observable in VLBI proper-motion measurements.
